

REPORT ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

QUESTIONNAIRE

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A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country: PERÚ
- Organization or entity responsible for the preparation of the report:
Name: Consejo Nacional de Ciencia, Tecnología e Innovación (CONCYTEC)
Website: www.concytec.gob.pe
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Please describe the role/mandate of your organization:
The National Council of Science, Technology, and Innovation (CONCYTEC) is a specialized technical body attached to the Presidency of the Council of Ministers. Its purpose is to regulate, direct, guide, promote, coordinate, supervise, and evaluate the State's actions in the field of science, technology, and innovation. Additionally, it seeks to promote and drive their development through coordinated and complementary action among the members of the National System of Science, Technology, and Innovation (SINACTI).
- Officially designated contact person who completed this survey:
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- Brief description of the consultation process established for the preparation of the report:
Deputy Directorate for Information and Knowledge Management maintains regular engagement with representatives of institutions within the National System of Science, Technology, and Innovation (SINACTI) as part of its functions.

B. Member States' questionnaire or report on the implementation of the 2021 Recommendation

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATE BENEFITS, AND CHALLENGES, AS WELL AS THE DIVERSE PATHS TO OPEN SCIENCE.

1.1. Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions, as well as affiliated organizations in your country?

YES

In December 2024, an official letter was sent to Universities and Public Research Institutes emphasizing the importance of complying with current regulations within the framework of strengthening the National Network of Information on Science, Technology and Innovation, in accordance with Law No. 31250 of the National Science, Technology and Innovation System (SINACTI); in particular, article 38 on the Promotion of Open Science and Law No. 30035, Law that regulates the National Digital Repository on Science, Technology and Innovation called ALICIA (the Spanish acronym for "Open Access to Scientific Information").

In February 2025, the UNESCO Recommendation on Open Science document was sent to the same institutions as a reference and complement to the previous official communication.

1.2. Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES

The Spanish version is available.

1.3. Have there been awareness raising activities on open science, including all its key elements, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: (i) 16.f, 16.g, 16.h)

YES

During 2024, the following awareness raising activities were carried out:

- Macro-regional meetings

Within the framework of the National STI Information Network, the macro-regional meetings have been held since 2021. In 2024, three meetings were held that brought together the member institutions of the National Network of Open Access Digital Repositories of Science,

Technology and Innovation (RENARE) and the institutions adhered to the PerúCRIS platform.

RENARE integrates and coordinates institutional open access repositories of more than 200 public and private institutions in the country, which are showcased in the ALICIA, a platform that centralizes and disseminates the country's scientific and academic production, promoting free access to knowledge. RENARE conducts annual evaluations of institutional repositories to ensure compliance with established guidelines.

PerúCRIS is a platform for the collection and dissemination of national activities related to science, technology and innovation, with more than 100 member institutions. It follows international standards such as OpenAIRE and the CERIF format.

The objective of the meetings was to review the scope of the Law of the National System of Science, Technology and Innovation (Law No. 31250) and its regulations, published in June 2024, with regard to the management of information in STI, the promotion of open science and compliance with the timely deposit in open repositories of research results with public funds, as well as to present the preliminary results of the evaluation of open institutional repositories of the 2024 call:

- Northern Macro-regional Meeting, was held on August 21 in the city of Chiclayo. The event was attended by 25 member institutions of the Network. The event was co-organized with the Universidad Católica Santo Toribio de Mogrovejo - USAT
- Southern Macro-regional Meeting, was held on November 13 in the city of Arequipa. The event was co-organized with the Universidad Nacional de San Agustín de Arequipa and representatives of 17 institutions participated
- Central Macro-regional Meeting, was held on November 27 in the city of Lima, the event was co-organized with the Universidad Femenina del Sagrado Corazón and representatives of 53 institutions participated.

- Research Data Management Workshops

Workshops were held to adopt good practices in research data management, aimed at researchers accredited in the National Scientific, Technological and Innovation Registry (RENACYT), where the need to incorporate the data management plan in project proposals and comply with the deposit of research data from publicly funded projects in open repositories was addressed.

During 2024, four workshops will be held in various cities of the country:

- Lima (August 12, Universidad Nacional Mayor de San Marcos),

- Chiclayo (August 20, Universidad Católica Santo Toribio de Mogrovejo)
 - Iquitos (August 23, Universidad Nacional de la Amazonía)
 - Arequipa (October 18, Universidad Católica Santa María).
- Resinfra Research Infrastructure Project

CONCYTEC participates in the Resinfra and Resinfra PLUS projects that seeks a sustainable partnership in research infrastructures between the European Union, Latin America and the Caribbean. Under this Horizon Europe program, the "Roadmap for the Construction of an EU-LAC Area" aims to identify the best infrastructures in Latin America and Europe to enhance their community use and impact on science and technology development. Additionally, work is underway to create a Latin American and Caribbean Research Infrastructure Forum (LACFRI) to facilitate coordination between open, unique, and large-scale infrastructures in the region.

On May 7-8, 2024, the 13th meeting of the EU-CELAC RI working group was held in Lima, presenting actions implemented in both regions, identifying infrastructures from Latin America, the Caribbean, and Europe.

From November 25-26, 2024, CONCYTEC, as the National Contact Point for the Horizon Europe Research Infrastructure Program, organized and participated in the event "Opportunities for Cooperation in the Horizon Europe Program" in the city of Arequipa.

During 2025, additional macro-regional meetings and workshops are planned, including:

- Meetings of the National STI Information Network
- Training sessions for research data management
- Open science promotion events

1.4. Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science (ii), in publicly funded research? (ref.; (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)

YES.

- Update of the ALICIA Guidelines

During 2025, the updated ALICIA 2.0.1 Guidelines will be published, further expanding the definition of open science values and principles in publicly funded research. The previous version (August 2020) already outlined the definition and scope of open science; however, this new edition incorporates specific and detailed guidelines at the metadata level. The goal is to enable open institutional repositories to integrate

both research data management plans and datasets into their collections in the short term.

- Incorporation of the Data Management Plan in the guidelines of the calls for publicly funded projects

Since 2025, calls for public funding under the National Program for Scientific Research and Advanced Studies (PROCIENCIA) have required the preparation and submission of a Data Management Plan as part of the research proposal requirements. This model is based on the Horizon Europe framework and fosters a strong commitment to the management and accessibility of research data.

- Macro-regional workshops 2024 and 2025

Regional workshops are organized for researchers and SINACTI entities to promote the dissemination of open science values and principles, as well as the processes for managing and depositing data and research outputs from publicly funded projects in open repositories.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

- 2.1. Does your country have a national policy, strategy or plan of action on science, technology and innovation (STI)?

YES

- Peru has a National Policy for the Development of Science, Technology, and Technological Innovation, which was approved in 2016, prior to UNESCO's recommendations on Open Science. A revised version of this policy is currently under development and is expected to be approved by the end of 2025.
- Although the current policy does not incorporate aspects of open science, in June 2024 the regulations of the Law of the National System of Science, Technology and Innovation (Law No. 31250) were published, which specifically establishes (Article 38) the Promotion of Open Science, based on its four key pillars: open scientific knowledge, open science infrastructures, open participation of social agents, and open dialogue with other knowledge systems.

- 2.2. In your country, is there a policy, or a set of policies and/or legal framework(s) (iv) at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (Ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.3, 16.h)

YES

Law No. 30035 establishes the National Digital Repository of Open Access Science, Technology, and Innovation, also known as ALICIA. Its purpose is to provide the academic, scientific, and general community with access to the intellectual heritage resulting from scientific, technological, and innovative production carried out in public sector entities or funded by the State. The law aims to preserve, safeguard, and ensure open access to these resources.

Law No. 31250, the Law of the National System of Science, Technology, and Innovation (SINACTI), was enacted in 2021 and regulated in June 2024. The law's regulations explicitly establish the Promotion of Open Science, based on its four key pillars: open scientific knowledge, open science infrastructures, open participation of social actors, and open dialogue with other knowledge systems. Additionally, it consolidates the National STI Information Network as an open scientific knowledge network, in alignment with Law No. 30035. Furthermore, it mandates the deposit of data management plans and research data from publicly funded projects in open repositories.

2.3. In your country, are there specific policy instruments (v) that aim to promote open science?

YES

Several instruments derived from the aforementioned legal frameworks are available, including:

- The National Network of Open Access Digital Repositories of Science, Technology, and Innovation (RENARE), established under the regulations of Law No. 30035, which defines the procedures and guidelines for repository operations and ensures open access to scientific production.
- ALICIA Directive No. 001-2020-CONCYTEC-P, which sets the guidelines, rules, and procedures for the integration and evaluation of RENARE's Institutional Repositories.
- The ALICIA Guide, a technical document that outlines the criteria for implementing and assessing institutional repositories, as well as the registration of metadata.
- Documents available at: <https://alicia.concytec.gob.pe/>

2.4. In your country, is there a national implementation plan, strategy, policy or roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)

UNDER DEVELOPMENT

- There are established legal frameworks, such as Law No. 31250, the Law of the National System of Science, Technology, and Innovation (SINACTI), which promotes Open Science, open

access repositories, and the open repository of publicly funded research data and products.

- Similarly, Law No. 30035 regulates the National Digital Repository of Open Access to Science, Technology, and Innovation, as well as the Network of Open Institutional Repositories.
- The implementation of the National STI Information Network and the STI Knowledge Management Platform is currently in progress. These initiatives aim to support the application of Open Science in accordance with the regulations of Law No. 31250 and Law No. 30035.

2.5. In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at the institutional level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.g)

YES

Law No. 30035 establishes the National Digital Repository of Open Access to Science, Technology, and Innovation (ALICIA), which integrates the National Network of Open Access Digital Repositories of Science, Technology, and Innovation (RENARE). Institutional open access repositories (currently over 200) are evaluated annually to ensure compliance with ALICIA guidelines.

Law No. 31250, which governs the National System of Science, Technology, and Innovation (SINACTI), establishes the following:

- o Promotion of Open Science through its four key pillars: open scientific knowledge, open science infrastructures, open participation of social actors, and open dialogue with other knowledge systems (Article 38).
- o Provision of research data (Article 39): Entities conducting research funded with public resources must make their research data publicly available in an open format, except in cases where exceptions are outlined by the governing body's directives.
- o Submission of a Data Management Plan: Individuals or organizations applying for public funding to support STI activities must submit a Data Management Plan during the application process.
- o Obligation to deposit research outputs: Individuals or organizations that have received public funding for STI-related activities must deposit both their data management plans and research outputs (such as research data, articles, manuscripts, book chapters, books, and other scientific publications) in institutional repositories, as required by Law No. 30035. If no institutional repository is available, the data must be deposited in CONCYTEC's institutional repository

2.6. In your country, are there specific funding mechanisms (vi) for open science at the national level? (ii) 17.j, (iii) 18.j, (iv) 19.c)

NO

There are no specific mechanisms in place yet. However, publicly funded research projects are required to deposit their corresponding research outputs and data in open access repositories.

2.7. In your country, have open science practices (vii) been integrated in existing research funding mechanisms?

YES

The regulations of Law No. 30035 mandate the deposit of research outputs that are partially or fully funded with public resources in open institutional repositories.

Additionally, the regulations of Law No. 31250 establish the following obligations (Article 39):

- SINACTI entities conducting research funded by public resources must make their research data publicly available in an open format, except in cases where exceptions are outlined by the governing body's directives.
- Individuals or organizations applying for public funding to support STI (Science, Technology, and Innovation) activities must submit a Data Management Plan as part of the application process.
- Individuals or organizations that have received public funding for STI-related activities must deposit both their data management plans and research outputs (including research data, articles, manuscripts, book chapters, books, and other scientific publications) in institutional repositories. If no institutional repository is available, the data must be deposited in CONCYTEC's institutional repository, as required by Law No. 30035.

2.8. Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (Ref.: 17.i)

NO

The feasibility of implementing public-private partnerships for this purpose has not yet been assessed.

2.9. Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy ? (ref.: (ii) 17.j, 23.c)

UNDER DEVELOPMENT

At the national level, the following strategic indicators are considered:

- Number of institutions (universities and Public Research Institutes) whose STI information or management systems are integrated with governing body platforms.
- Number of articles published with full-text open access.
- Number of publications for which research data is available in a public repository.

3. INVESTING IN OPEN SCIENTIFIC INFRASTRUCTURES AND SERVICES

- 3.1. What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditures? Please also indicate the year of publication of this data, and to which year(s) it refers. (Ref.: (III) 18.a)

Research and development expenditures as a percentage of national GDP:

- Year 2021: 0.14%
- Year 2022: 0.16%
- Year 2023: 0.18%

Source: STI Indicators Portal - CONCYTEC.

(<https://portal.concytec.gob.pe/indicadores/principales/>)

- 3.2. What is the latest official data for the percentage of your country's population that has access to reliable Internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

Percentage of the country's population aged 6 and over who have access to the Internet: 77.7%

Source: National Institute of Statistics and Informatics (INEI) in its technical report Statistics on Information and Communication Technologies in Households, as of the first quarter of 2024.

(<https://cdn.www.gob.pe/uploads/document/file/6543264/5702640-las-tecnologias-de-informacion-y-comunicacion-en-los-hogares-ene-feb-mar-2024.pdf?v=1719522078>)

- 3.3. Are there national research and education networks (NRNs) (viii) active in the country? (ref.: (iii) 18.c)

NO

Currently, there are no active National Research and Education Networks (NRENs) in Peru. However, initiatives are underway to establish them.

- 3.4. Are there national or institutional open science infrastructures (ix) in your country? (ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)

UNDER DEVELOPMENT

At the national level, the following initiatives are in place:

Spatial Data Infrastructure of Peru (IDEP): An open data platform that facilitates the production, use, and access to geographic information, maintained by the same public and private entities that generate geographic data (nodes). (<https://www.gob.pe/idep>)

National Network of Open Access Digital Repositories of Science, Technology, and Innovation (RENARE): Comprising more than 200 document repositories from public and private institutions, RENARE centralizes its metadata in the National Digital Repository of Open Access to Science, Technology, and Innovation (ALICIA). (<https://alicia.concytec.gob.pe>).

PeruCRIS Platform: A system that centralizes metadata related to Science, Technology, and Innovation (STI) activities from public and private entities across the country (<https://www.perucris.pe>).

SciELO Peru (Scientific Electronic Library Online): A platform providing full-text access to a collection of previously qualified Peruvian scientific journals, as part of a broader regional initiative.

Research Data Portal: A national network of research data repositories is planned for 2025, along with the implementation of a centralized repository to store research data from publicly funded projects.

- 3.5. Does your country host or fund any federated regional or international IT infrastructure for open science? (ref.: 18.e, 18.f)

NO

As a country, Peru does not host any regional or global infrastructure. However, some local platforms are integrated into regional networks. For example, the national open access repository ALICIA is connected to La Referencia, a platform that aggregates Open Science repositories across Latin America and Spain, recognized as a regional public good. Similarly, SciELO Peru is part of the SciELO Latin American Regional Network.

- 3.6. Is your country involved in any North-South, North-South-South and South-South collaboration to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref: (iii) 18.g)

YES

The Network of Institutional Open Access Repositories (RENARE) and the ALICIA National Repository participate in La Referencia, a regional Open

Science platform that connects repositories across Latin America and Spain.

The SciELO Peru open-access virtual library is part of the SciELO Latin American Network.

CONCYTEC is involved in the Horizon Europe Resinfra and Resinfra PLUS projects, which aim to establish sustainable partnerships in research infrastructures between the European Union, Latin America, and the Caribbean.

4. INVEST IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE.

- 4.1. Has systematic capacity building on open science taken in higher education and/or research performing institutions in your country? (Ref.: (IV) 19.a, 19.c)

NO

So far, only virtual training sessions and face-to-face meetings have been conducted, primarily involving vice-rectors for research from public and private universities. However, a systematic and broader capacity-building initiative has not yet been implemented.

- 4.2. Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

NO

So far, only virtual training sessions and face-to-face meetings have been conducted, primarily involving vice-rectors for research from public and private universities. However, a specific program for policymakers has not yet been implemented.

- 4.3. Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (Ref.: (IV) 19.b)

NO

Not yet

- 4.4. Are open educational resources (x), as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

NO

Not yet

- 4.5. Have there been any initiatives in your country to support science communication in line with open science values and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the general public at large? (Ref.: (IV) 19.e)

YES

In 2024, CONCYTEC organized virtual seminars and regional face-to-face meetings focused on educating researchers about various aspects of open science.

Additionally, the College of Librarians of Peru and the Open Access Peru Collective held the "Open Access Week Peru 2024" in October 2024, promoting the values and principles of open science to the broader community.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1. Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: (v) 20.b, 20.c, 20i)

NO

Not yet, however, a redesign of the criteria and evaluation processes for researchers to qualify to the National Scientific, Technological and Innovation Registry (RENACYT) is planned for 2025.

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025?

NO

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (Ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20.h)

NO

- 5.4 Have there been any unintended negative consequences (xii) of open science practices been reported in your country? (ref.: (v) 20)

NO

2. PROMOTING INNOVATIVE APPROACHES TO OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

YES, at the national level.

Promotion of pre-prints: Efforts are underway to promote the management of pre-publications on the SciELO platform for national open-access journals.

3. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science, as outlined in the 2021 Recommendation on Open Science? (Ref.: (VI) 22.a)

CONCYTEC actively participates in the Resinfra and Resinfra PLUS projects, which aim to establish sustainable partnerships in research infrastructures between the European Union, Latin America, and the Caribbean.

As part of this Horizon Europe initiative, the "Roadmap for the Construction of an EU-LAC Area" has been developed to identify and enhance the best research infrastructures in Latin America and Europe, fostering community access and maximizing their impact on scientific and technological development.

Additionally, efforts are underway to create the Latin American and Caribbean Forum on Research Infrastructures (LACFRI), which will facilitate coordination among open, unique, and large-scale research infrastructures across the region.

- 7.2 Does your country have a strategy or plan to stimulating cross-border multi-stakeholder collaboration in open science, including with regards to exchange of best practices, capacity building and metrics for open science? (Ref.: (vii) 22.b, 22.c, 22.f)

NO

- 7.3 Is your country involved in any discussions on the establishment of regional and international funding mechanisms for open science? (ref.: <https://unesdoc.unesco.org/ark:/48223/pf0000379949/PDF/37994eng.pdf.multi.nameddest=20/p.nameddest=20/.pag=21>)

NO

4. **GENERAL CONSIDERATIONS**

8.1 Are there specific external causes that may prevent the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

NO